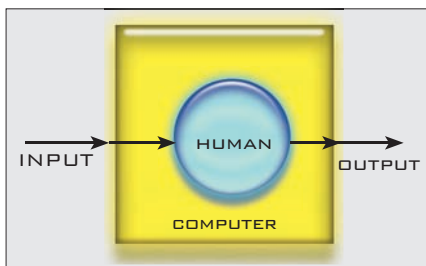


Self-dusting solar panels

Scientists at the Boston University, in collaboration with NASA, have developed sensors that remove dust from solar panels

Cyber-police station at Kolkata

Kolkata police has launched a Cyber-Police station in association with Infosys to deal with cyber-related crimes



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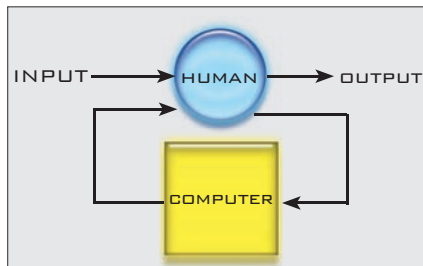
As in, user will be doing something else at the same time along with computing. Thus, the computer should serve as an augment to the user's intellect and senses.

As input, different techniques are followed as per the use of the device. Many use input methods like gesture, speech recognition and context awareness. Some of them use "Kiers" and track balls as input device. 'Kiers' are key switches mounted to a grip rather than to a board, as with a keyboard.

Research

The vast opportunities on offer in the wearable computing segment has pioneered the imaginations of designers and researchers in a wide variety of fields. Many major Institutes like MIT have started research in this field. The scientists at MIT came up with some interesting projects such as SNiF-Social Networking on Fur, Reality Mining Cell Phone Project.

Reality Mining Project is one of the largest mobile phone projects attempted. The research takes into consideration, the increasingly widespread use of mobiles to provide insight into dynamics



Augmentation

of both individual and group behaviour. This project focuses on the information flow between people and how can they change a group's interaction to promote better functioning.

"Smart Clothes and Wearable Technology Research Group"(SCWT) is a research group set up in collaboration between the University of Wales, Newport and the National Centre for Product Design and Development Research (PDR). They aim to create 'smart' clothes for the future. The project was launched in November 2004 and since then the group focuses on the role of wearable technology and how it can be deployed to create commercial possibilities in areas such as corporate wear and sports wear. They are working on clothes with embedded technologies with potential to enhance the quality of life for the elderly population.

Microsoft's SkinPut

Microsoft is one company which is showing keen interest in this field. The research going on is quite remarkable as quoted by many. Desney Tan, senior researcher in the Visualization and Interaction for Business and Entertainment (VIBE) group at Microsoft Research Redmond came up with something really interesting - SkinPut.

SkinPut seeks to move beyond the mouse

and physical keyboard, letting people communicate with their gadgets by gesturing, using sign language such as tapping their fingers.

Just imagine your iPhone's keyboard being as big as your arm and you wouldn't have to touch the keys at all. That's what SkinPut is all about. It aims to replace the physical modes of input like keyboards completely with the human body.

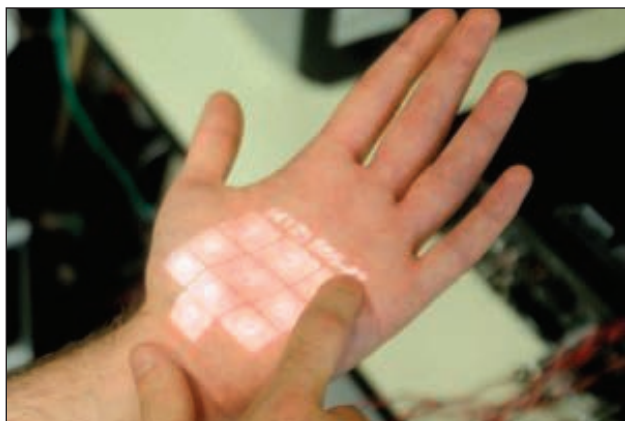
"Human beings are well evolved to communicate with their environments and other people in amazingly complex



SkinPut also supports gestures

ways," says Desney Tan. "Unfortunately, most of our current computer input devices, such as mice and keyboards, use very little of this bandwidth and provide a communication channel that could be much richer." SkinPut is a technology that appropriates the human body for acoustic transmission, allowing the skin to be used as an input surface. SkinPut takes as input, the finger-taps on the arm and the hand by analysing mechanical vibrations that propagate through the body.

This is done in two parts. One uses Electromyography, sensing of electrical muscle activity which can recognise finger gestures. The other one uses bio-acoustic sensors that can detect transmissions through the body. This would transform the human body into a tap-based input device. The bio-acoustic sensor can recognise acoustically dis-



A SkinPut prototype in use